PROJECT EVIDENCE REPORT

# GUVI Project 3

## AWS Infrastructure Automation

Terraform + GitHub Actions + AWS OIDC

### Deployment pipeline completed successfully

Captured evidence includes a successful Plan and Apply run, a working Nginx page, networking, S3 storage, Terraform remote state, and IAM configuration.

**Repository**

https://github.com/naviktg/Guvi-Project-3

**Application URL at evidence capture**

http://13.205.240.3

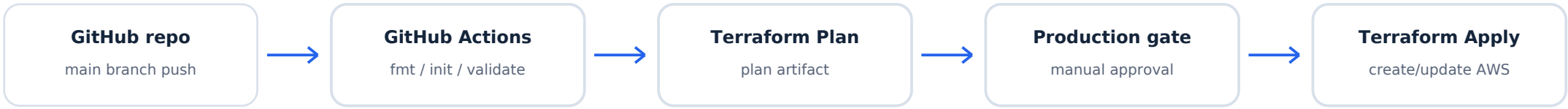
**Operational note**

The screenshots document the successful deployment before cost cleanup. After capturing the evidence, the EC2 instances were stopped and the Elastic IP was released. The address above is historical evidence and may no longer be reachable.

01 / OVERVIEW

# Project scope and deployment flow

This project provisions a small AWS web-server environment with Terraform and uses GitHub Actions for an automated, approval-gated deployment. Authentication from GitHub Actions to AWS uses OpenID Connect (OIDC), rather than storing long-lived AWS access keys in the workflow.

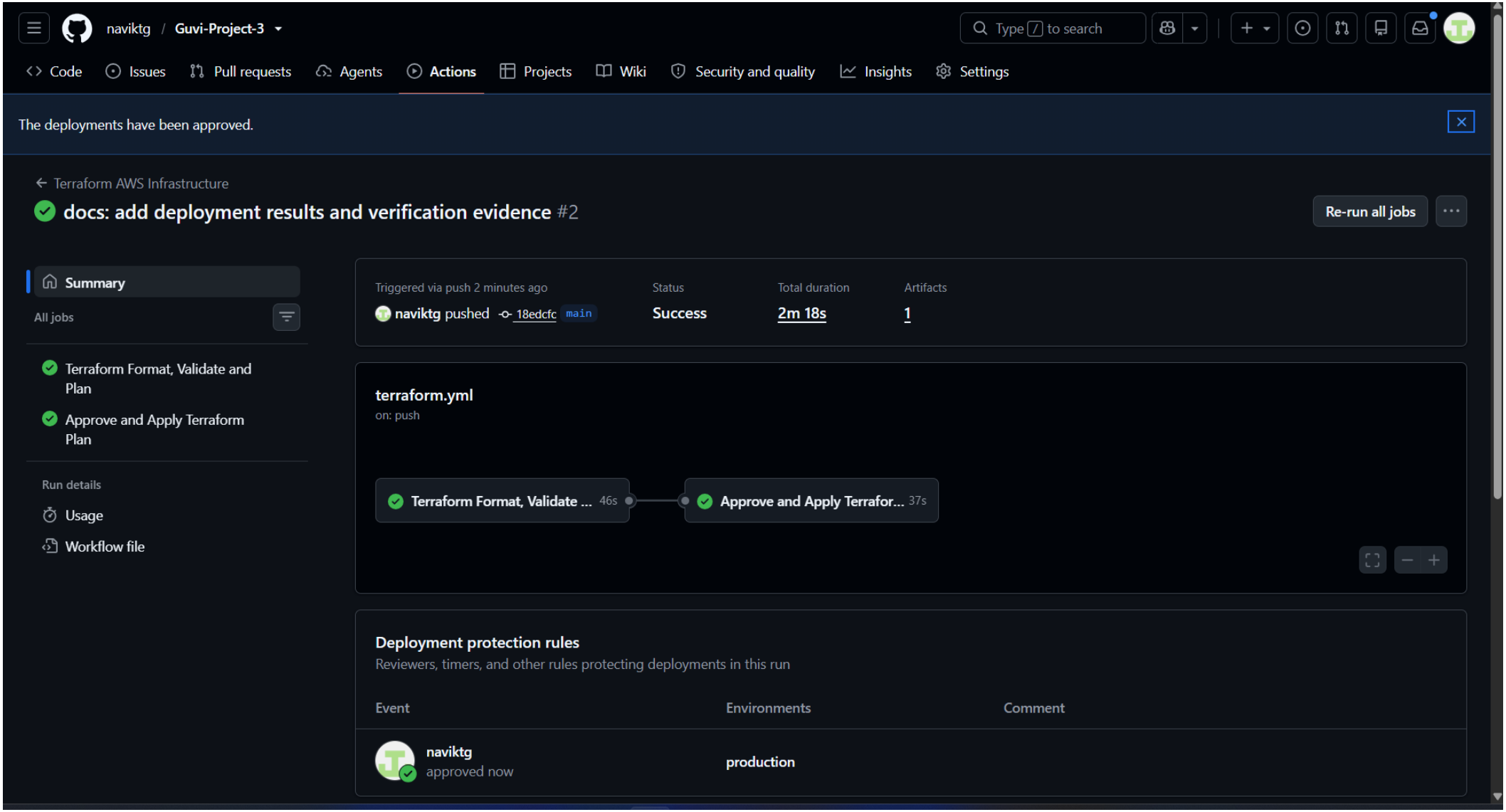



## Provisioned and verified components

- Networking: VPC 10.0.0.0/16, two public subnets in ap-south-1a and ap-south-1b, Internet Gateway, route table.
- Compute: EC2 t3.micro, Nginx web server, Elastic IP at evidence capture, web Security Group.
- Storage: application S3 bucket with versioning, SSE-S3 encryption, public-access blocking; S3 remote state.
- Identity: EC2 IAM role and instance profile with an application S3 read policy.
- Pipeline: GitHub Actions Plan artifact, production approval, Terraform Apply, output and web validation.
- Repository: naviktg/Guvi-Project-3 (source and workflow history).

# GitHub Actions: successful Plan and Apply

The successful workflow run shows both the planning job and the approval-gated apply job completed, with one plan artifact.



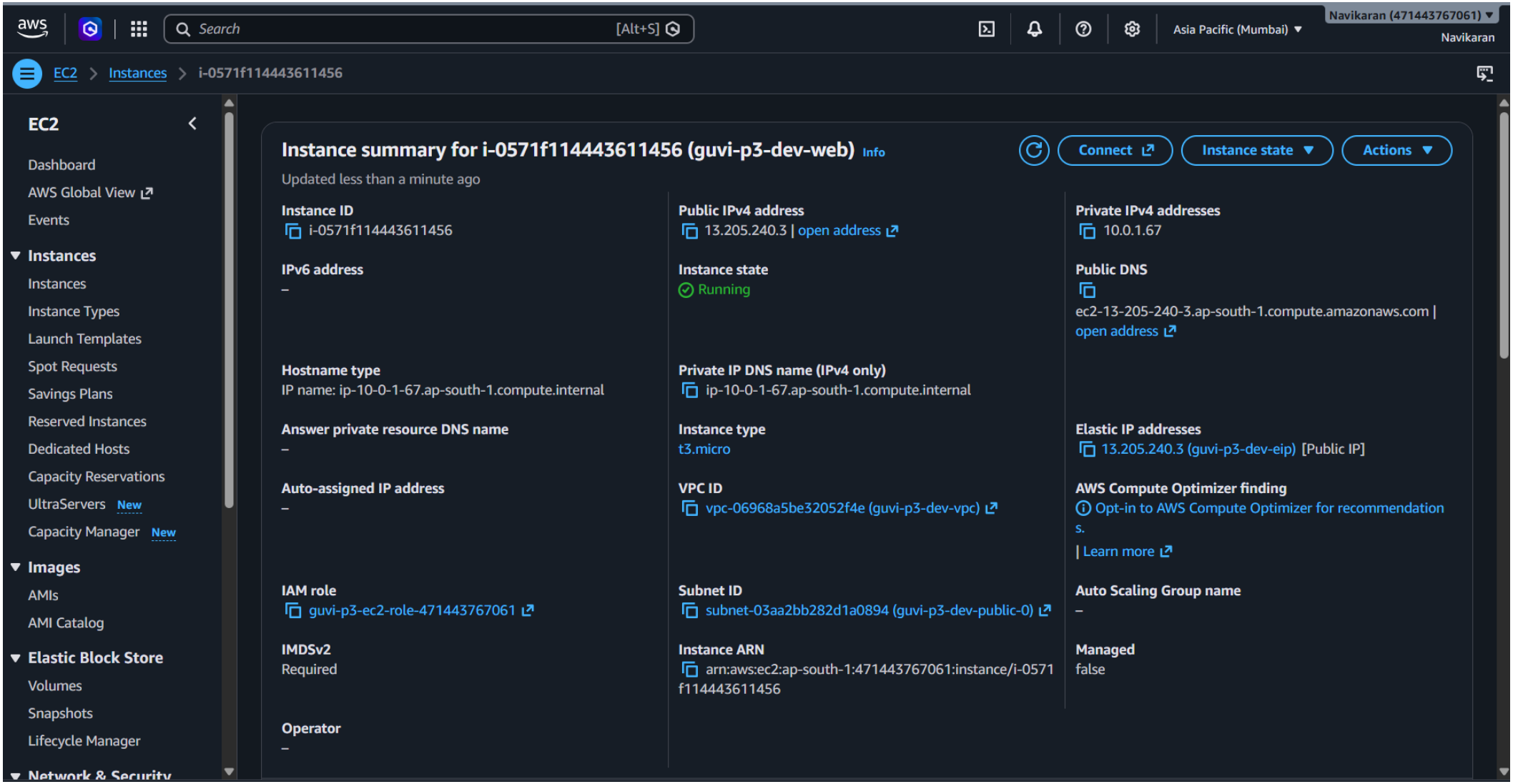
# Nginx website opened in a browser

Browser verification captured while the application was reachable at the recorded Elastic IP.



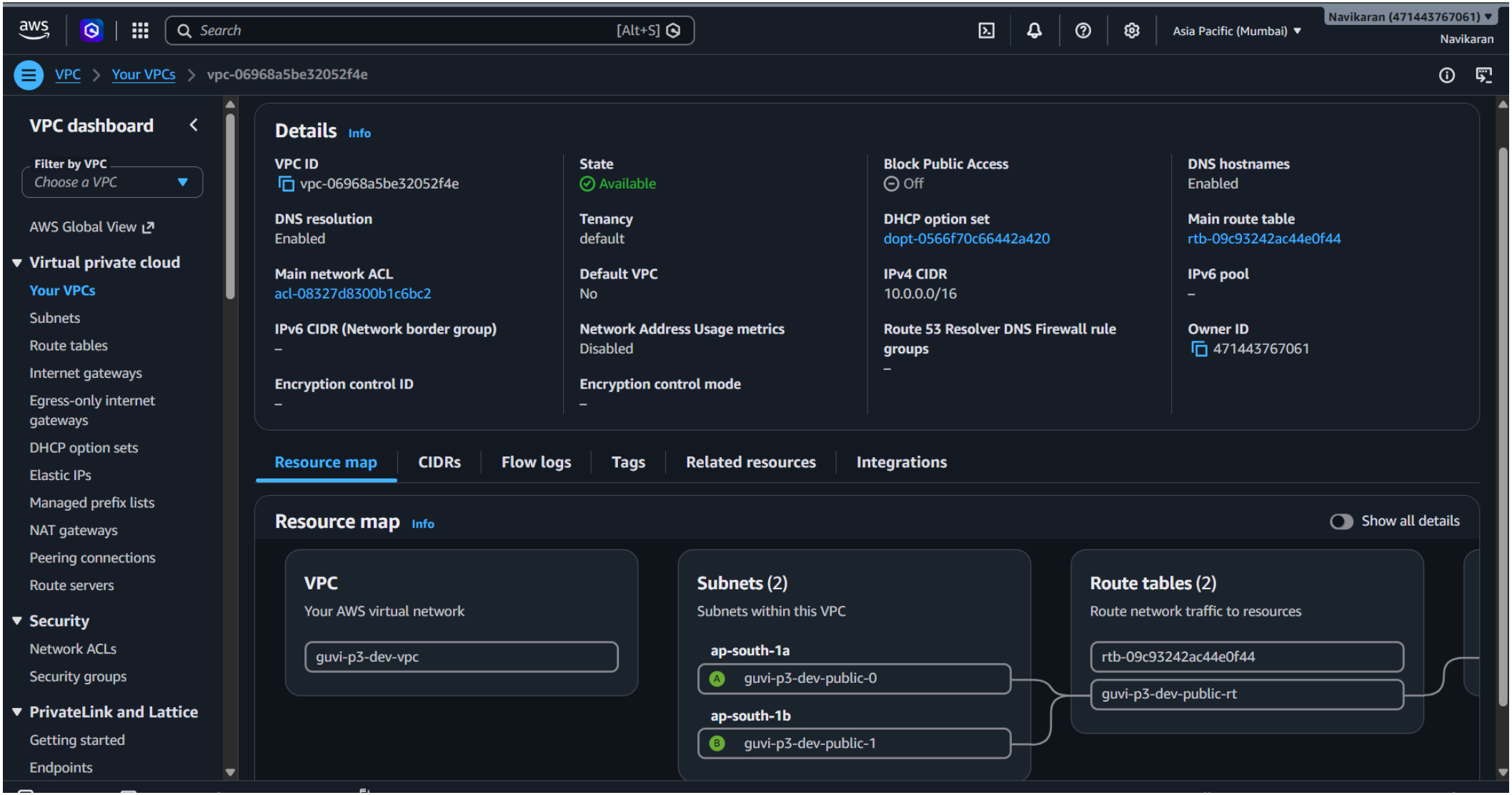
# EC2 instance running with Elastic IP and IAM role

Instance summary confirms the t3.micro instance was running, with public IPv4 13.205.240.3 and the EC2 role attached at evidence capture.



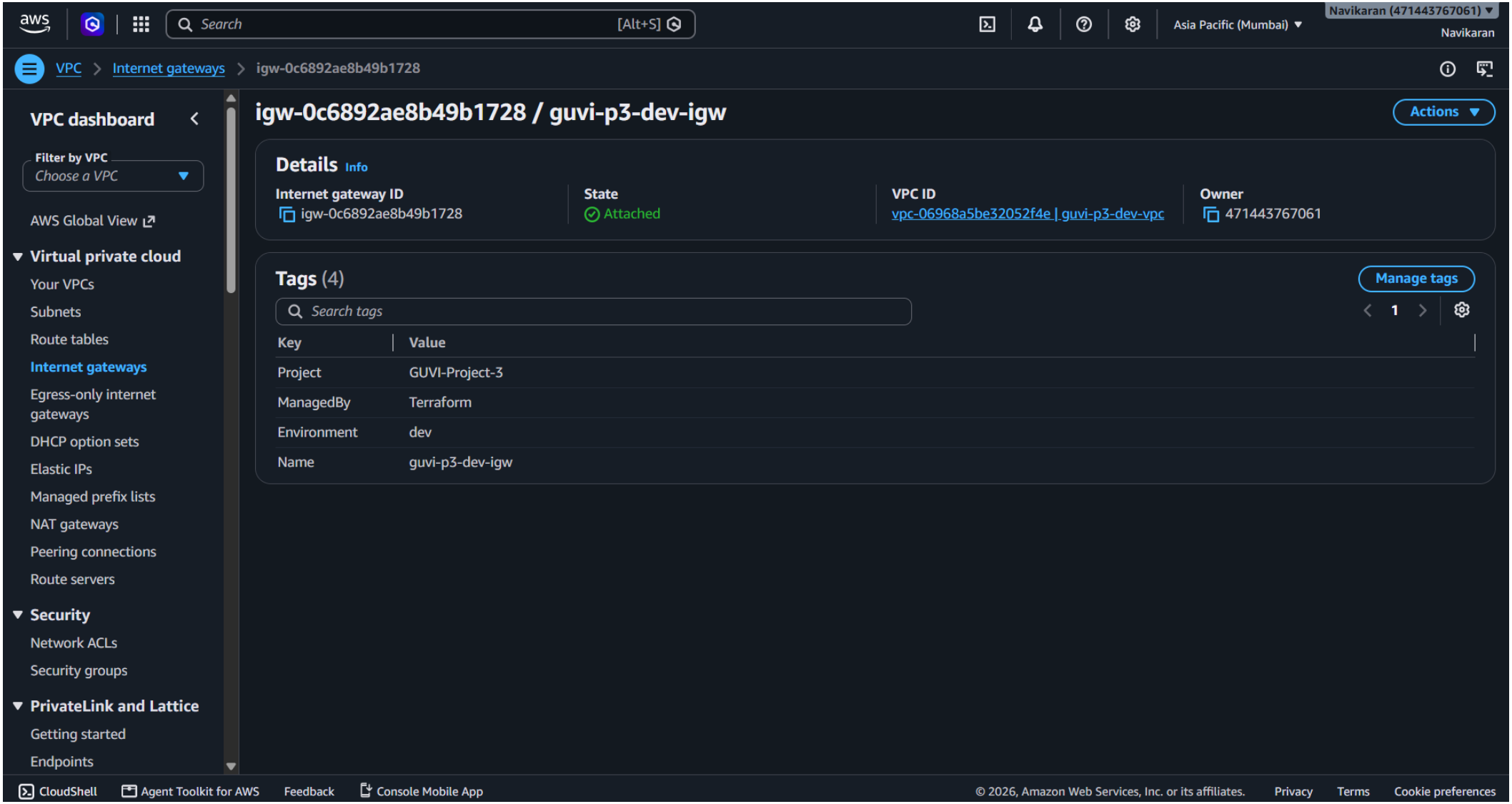
# VPC resource map and two public subnets

VPC CIDR 10.0.0.0/16 and two public subnets are visible in the AWS resource map.



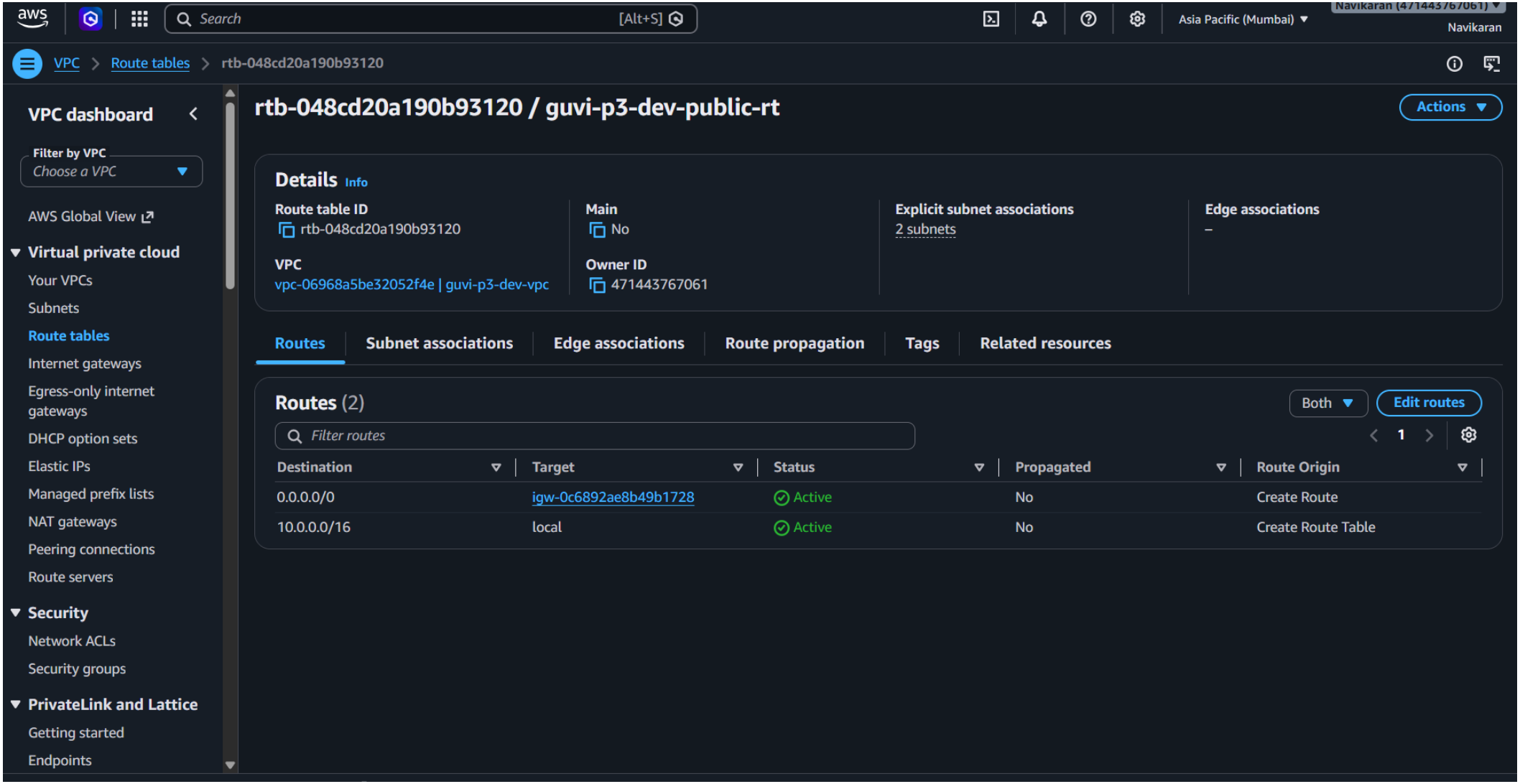
# Internet Gateway attached to the project VPC

The Internet Gateway status is Attached to the project VPC.



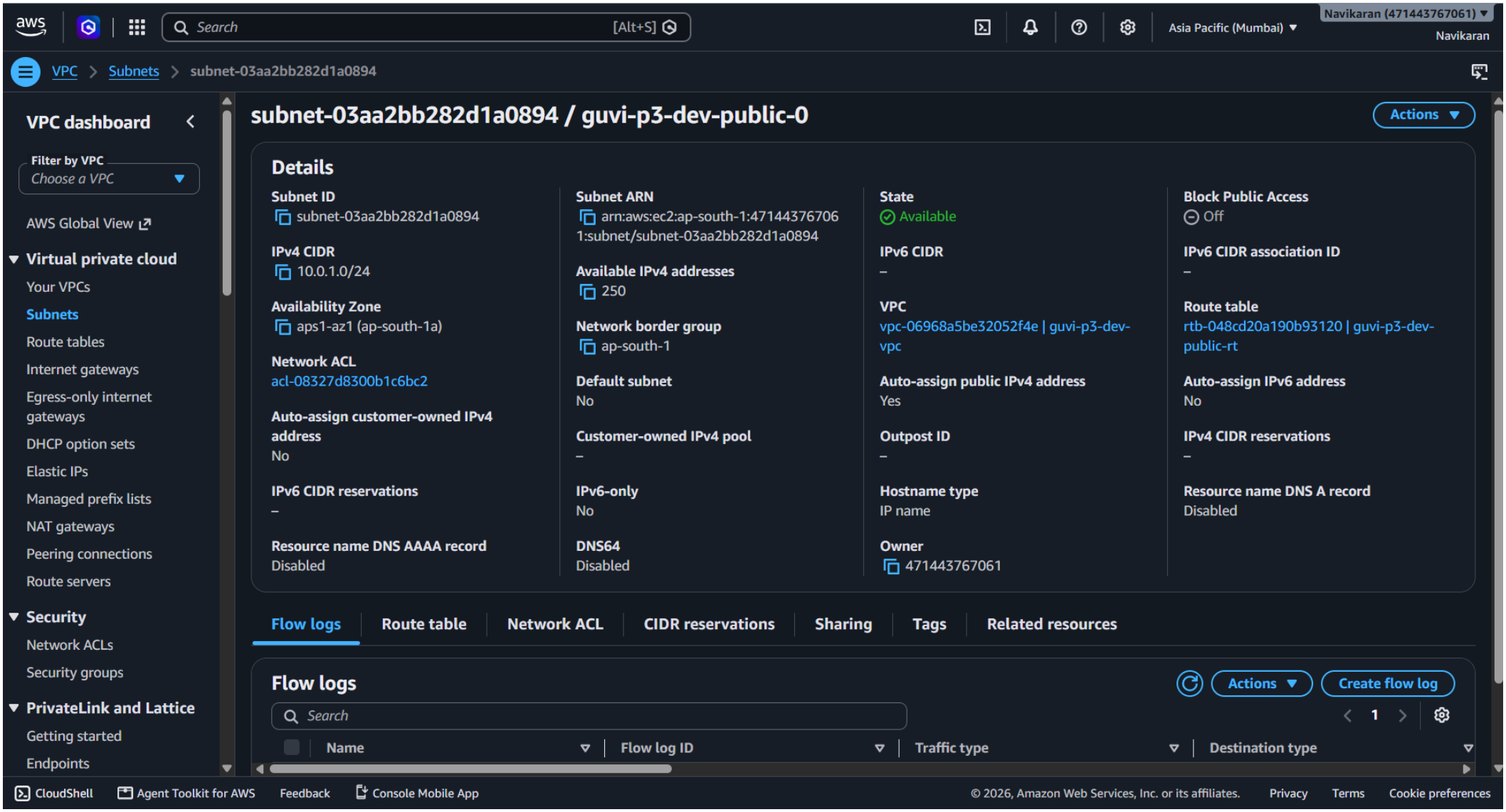
# Public route table with active Internet Gateway route

The route table contains 0.0.0.0/0 to the Internet Gateway and the local 10.0.0.0/16 route; both are Active.



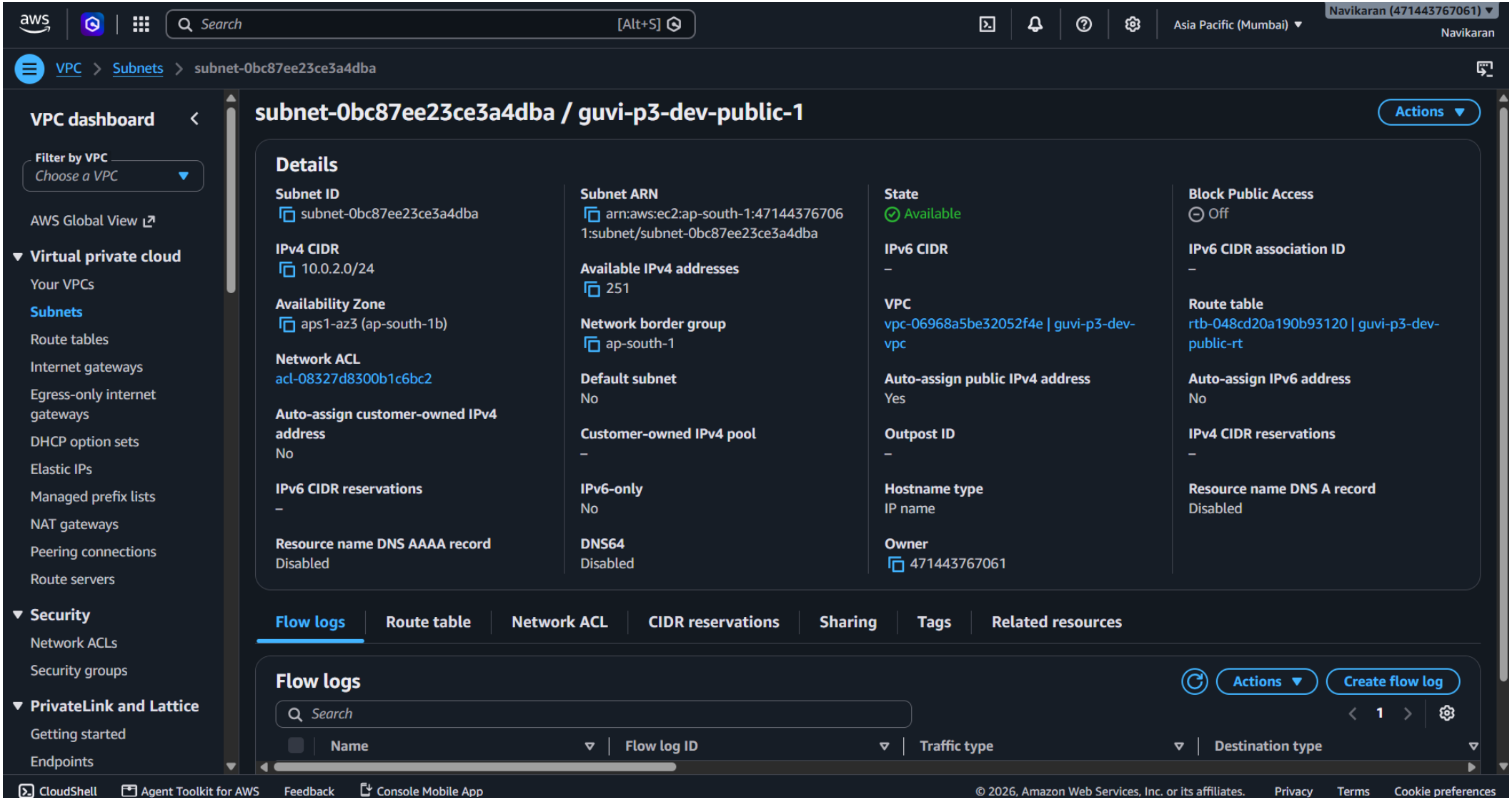
# Public subnet 1 details

Subnet guvi-p3-dev-public-0 uses 10.0.1.0/24 in ap-south-1a and is associated with the public route table.



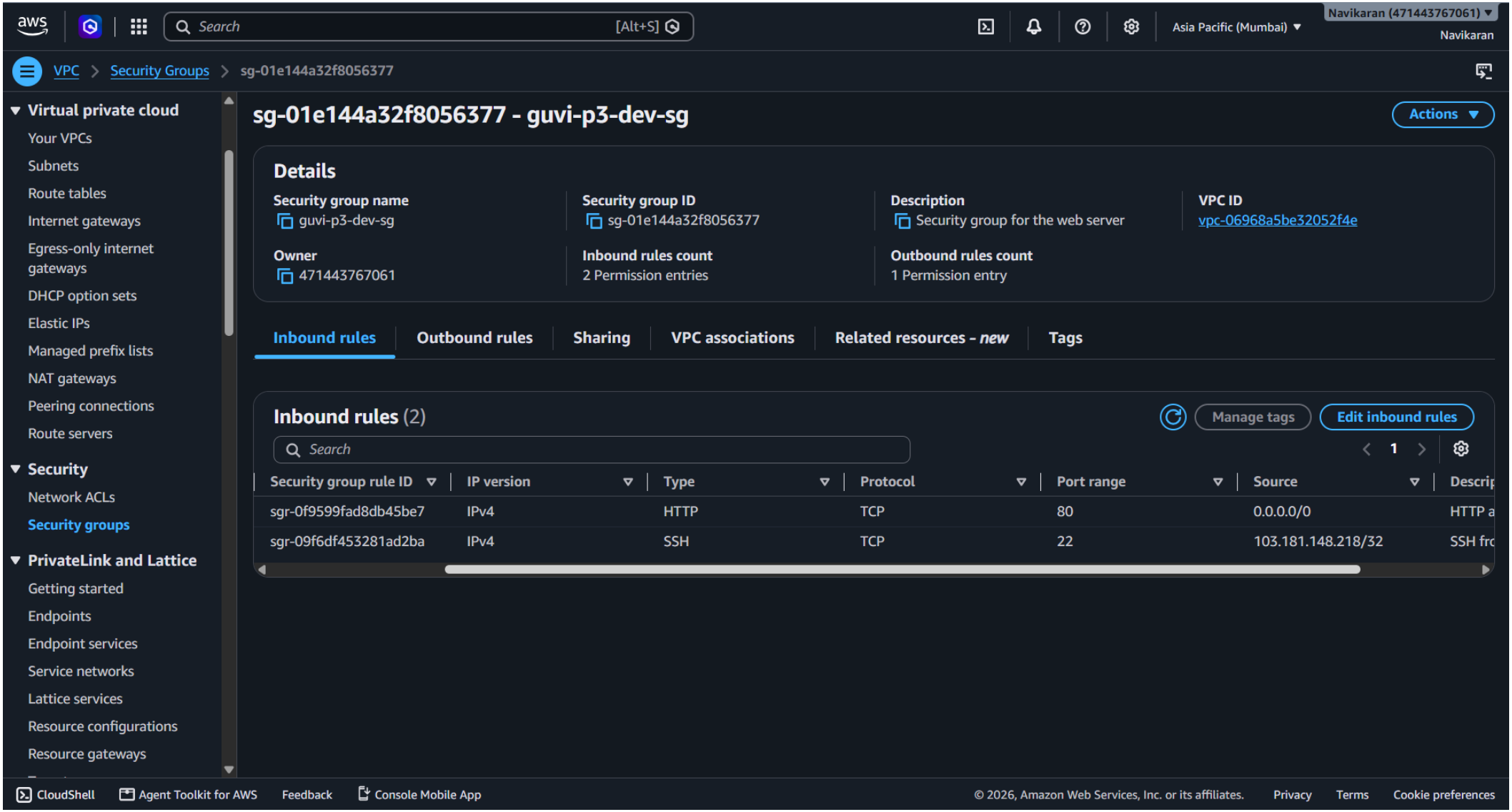
# Public subnet 2 details

Subnet guvi-p3-dev-public-1 uses 10.0.2.0/24 in ap-south-1b and is associated with the same public route table.



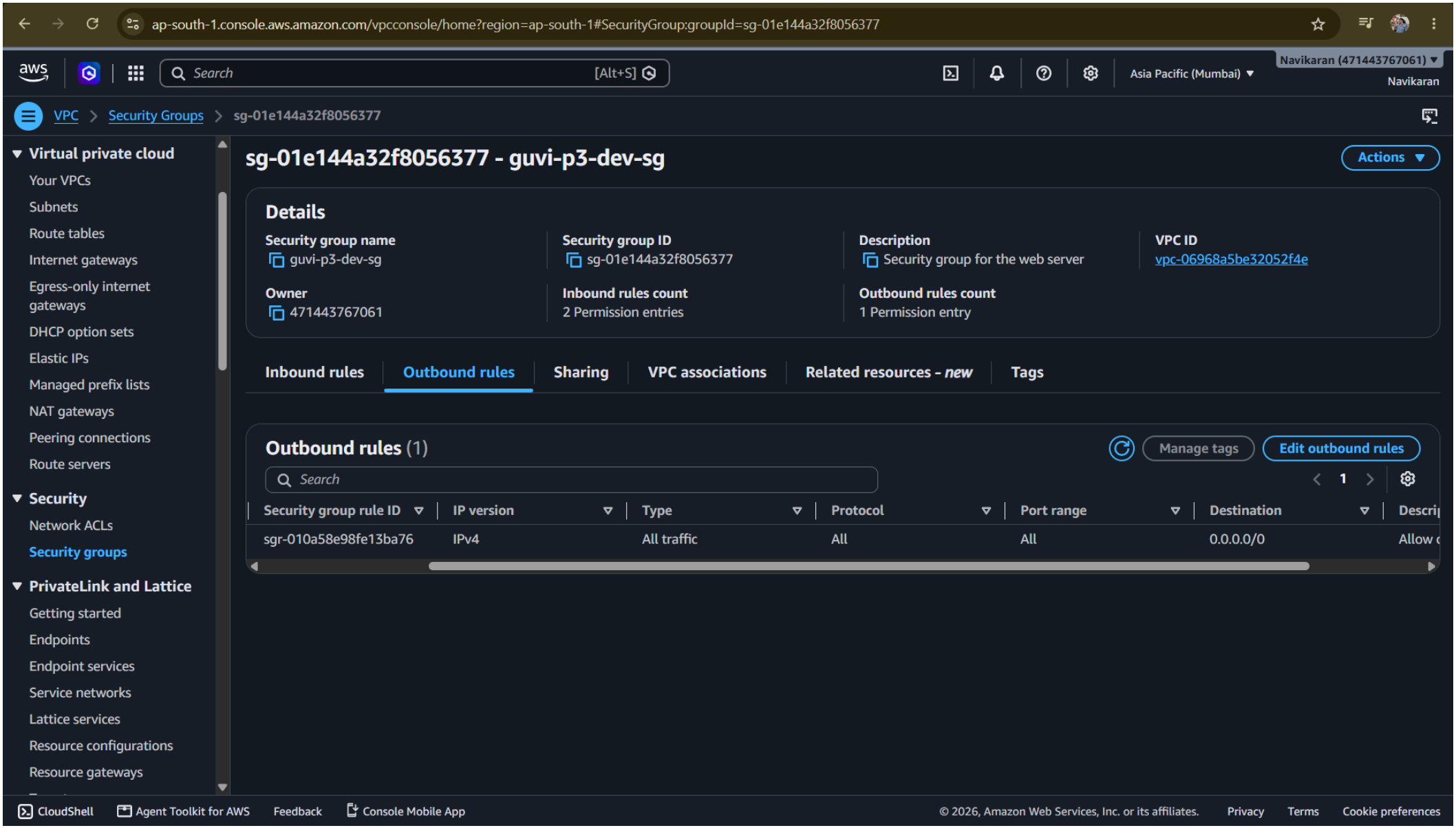
# Security Group inbound rules

HTTP port 80 is open to 0.0.0.0/0; SSH port 22 is restricted to the configured /32 source address.



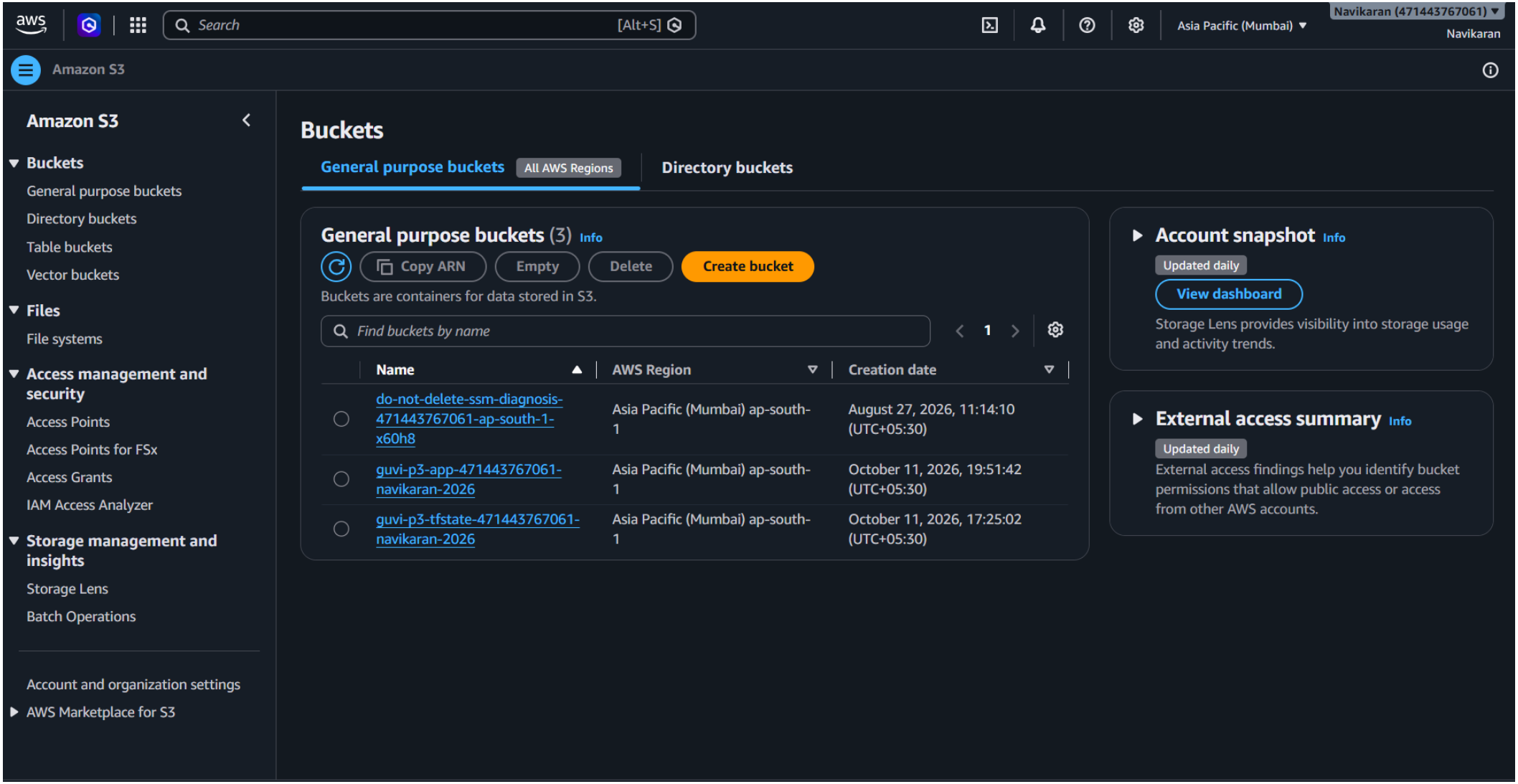
# Security Group outbound rules

The Security Group has one outbound rule allowing all traffic to 0.0.0.0/0.



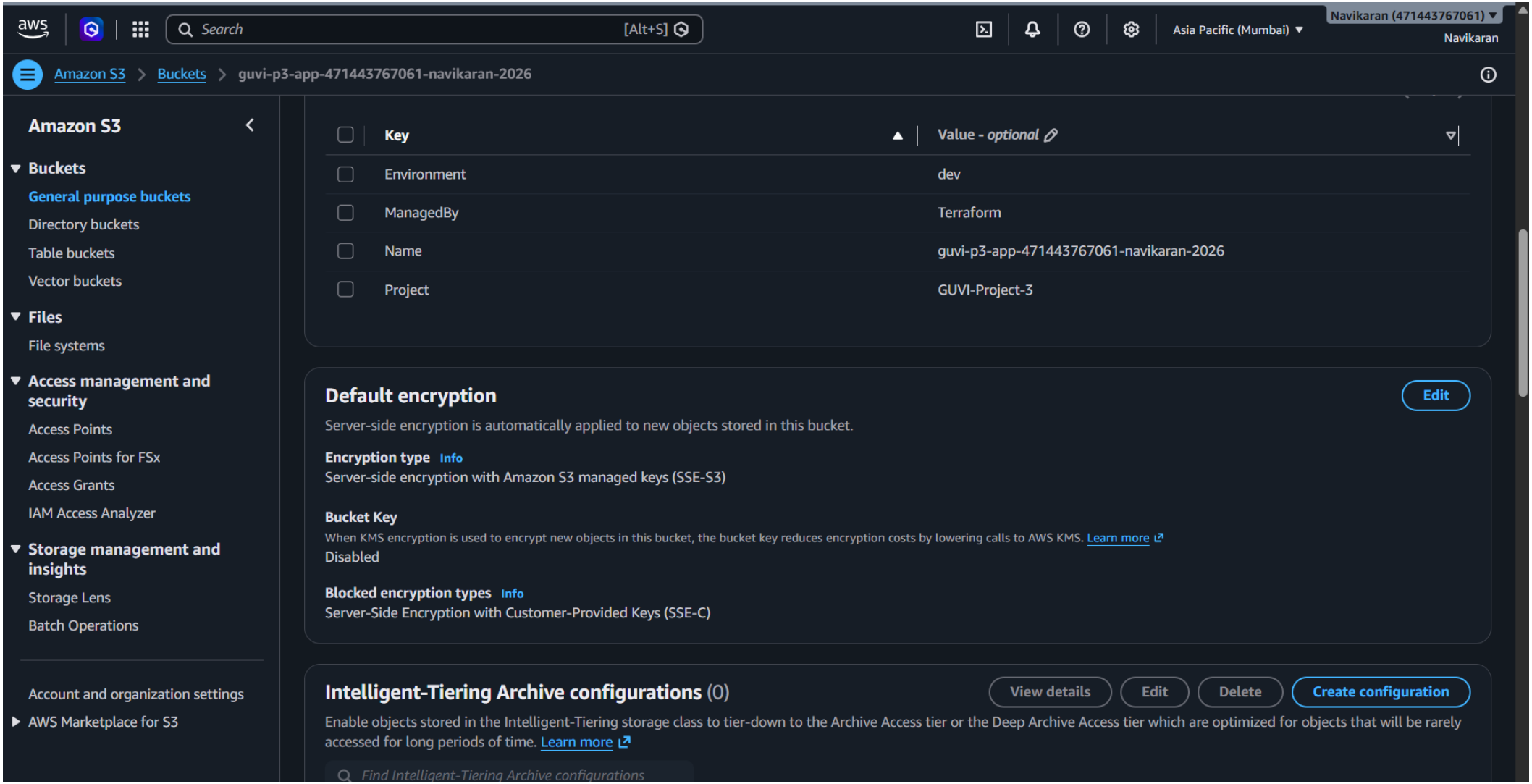
# S3 bucket inventory

The application bucket and the Terraform state bucket are both present in the S3 bucket list.



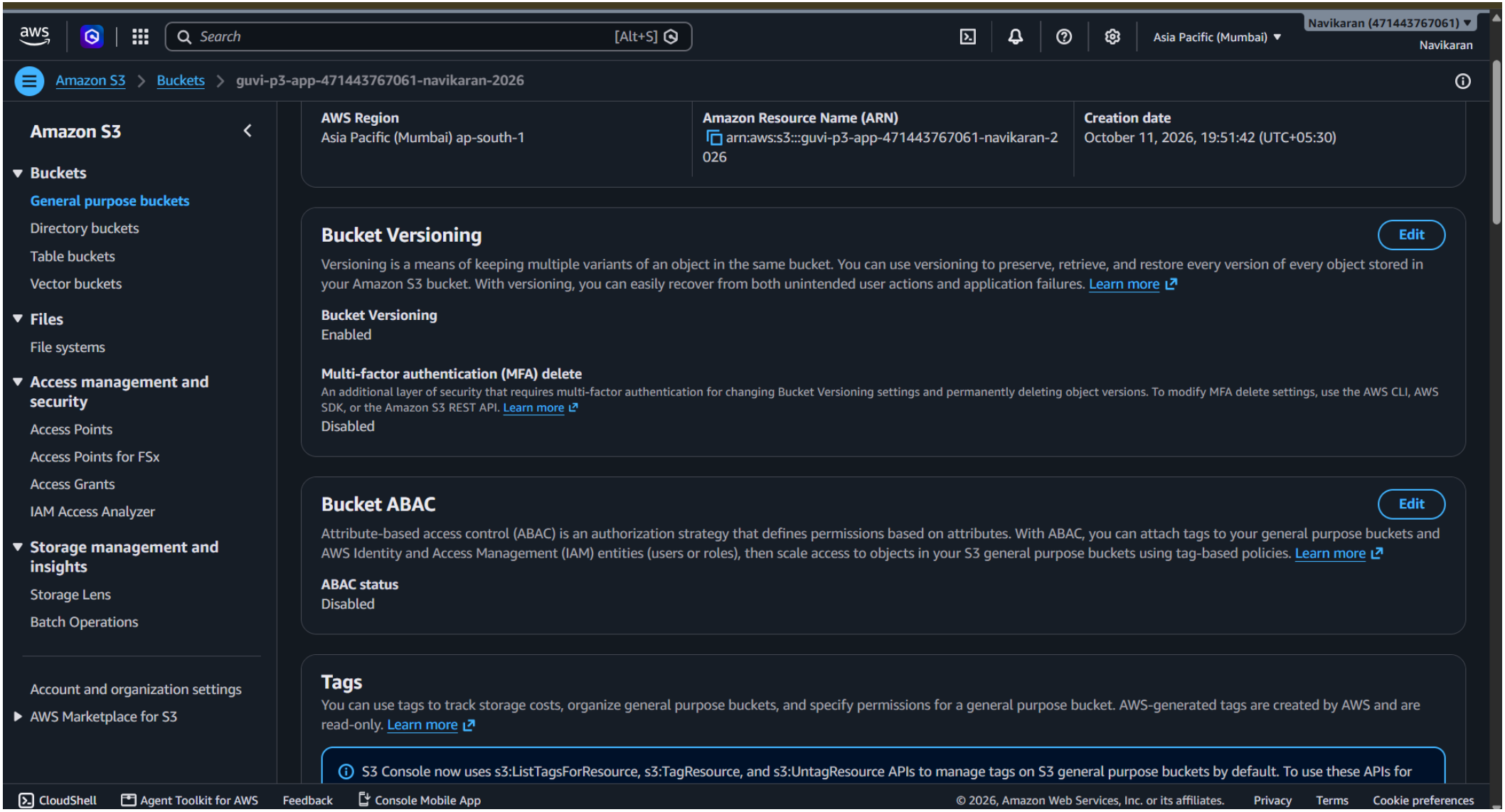
# Application bucket default encryption

The application bucket shows server-side encryption with Amazon S3 managed keys (SSE-S3).



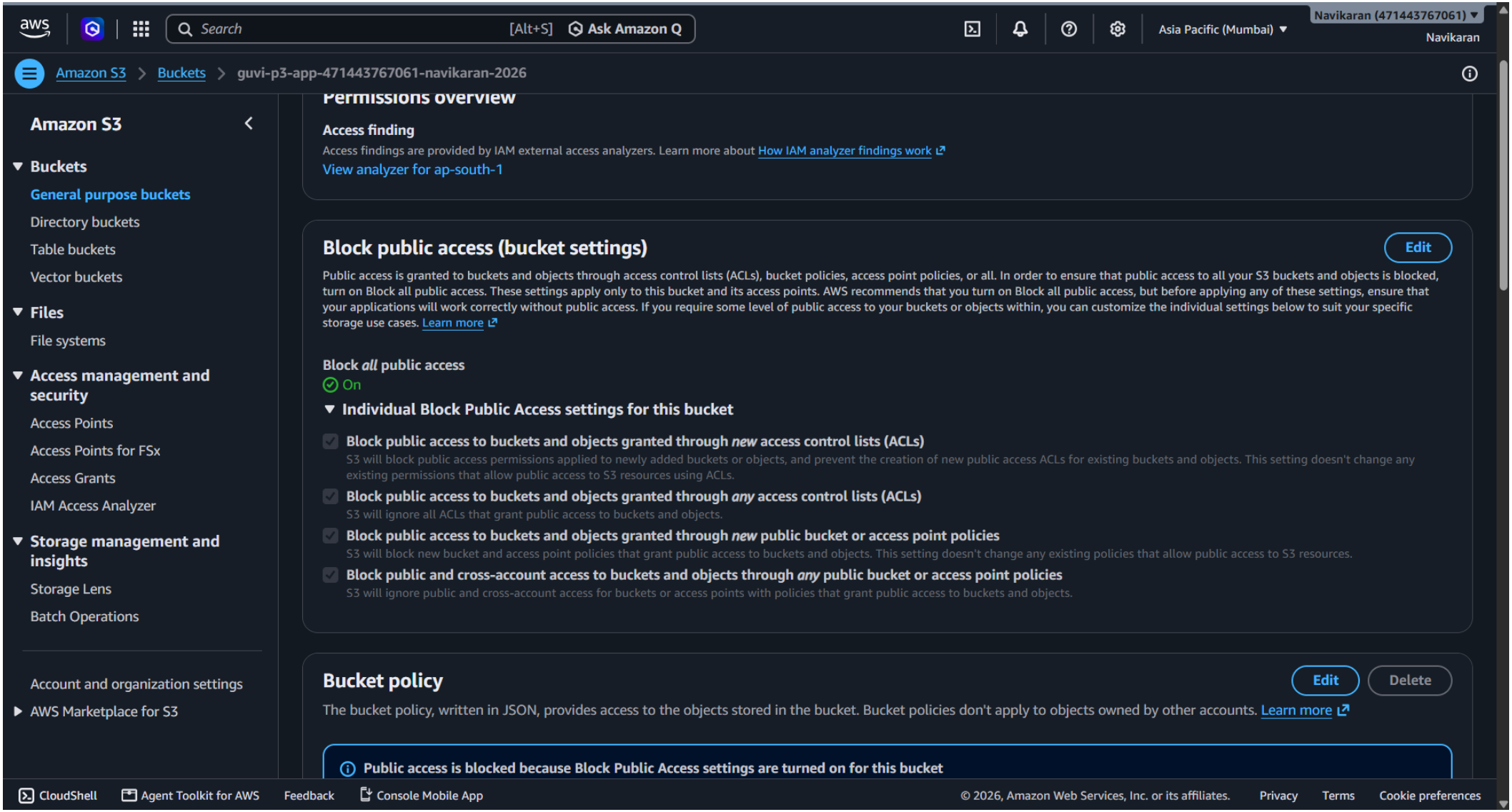
# Application bucket versioning

Bucket versioning is shown as Enabled.



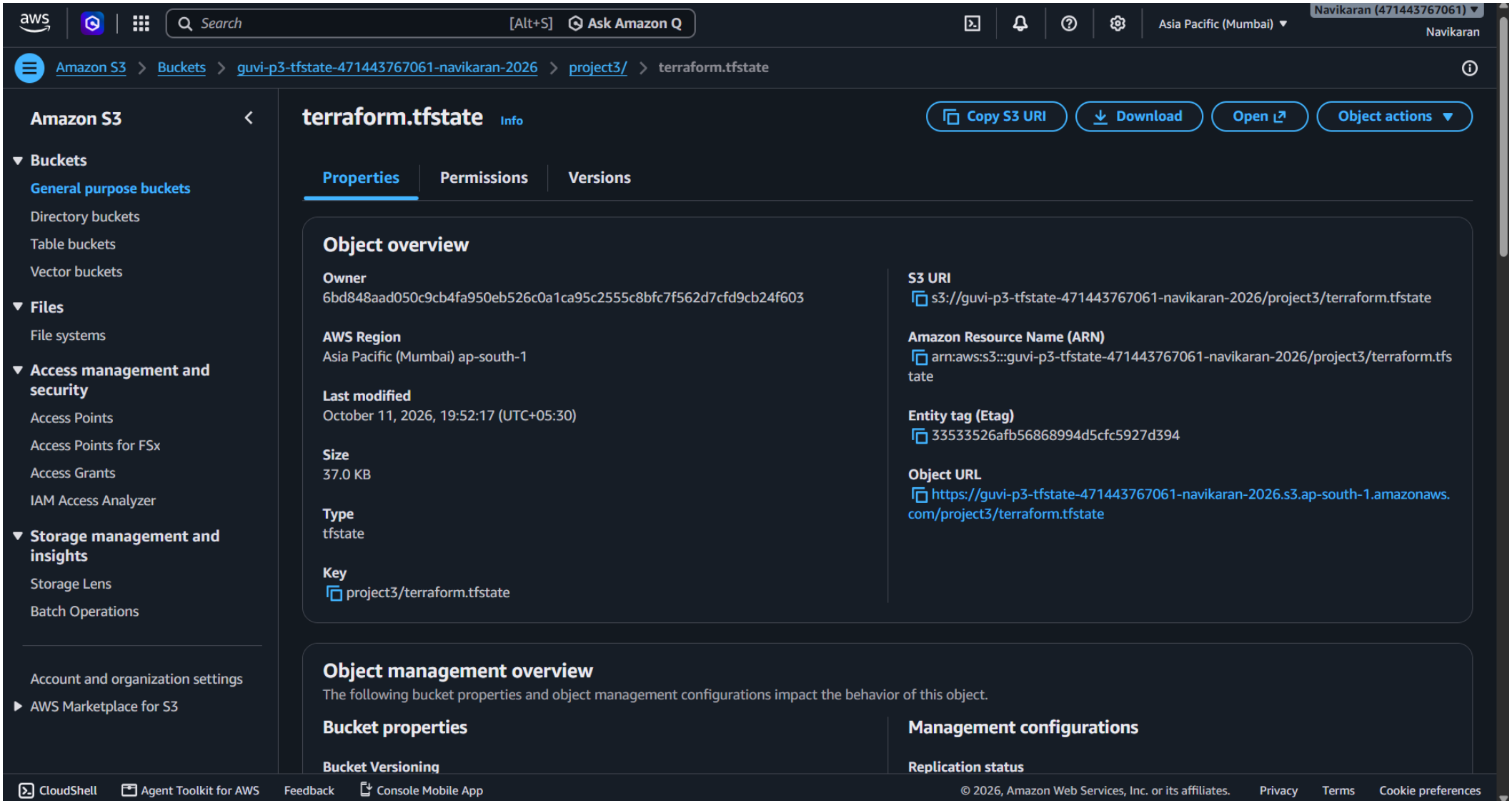
# Application bucket public-access protection

Block all public access is On, and the individual public-access block settings are enabled.



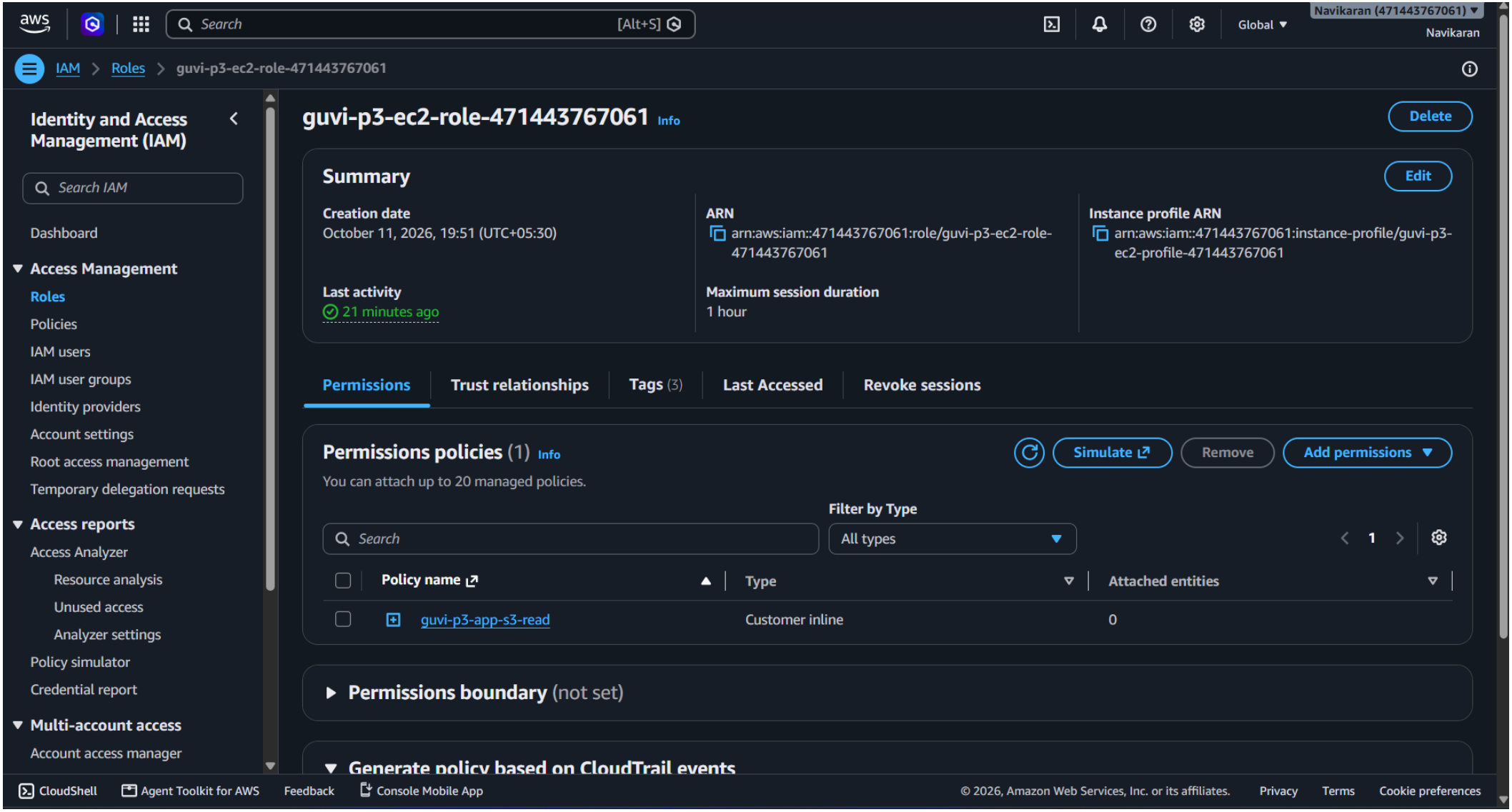
# Terraform remote state object in S3

The remote Terraform state object exists at project3/terraform.tfstate in the dedicated state bucket.



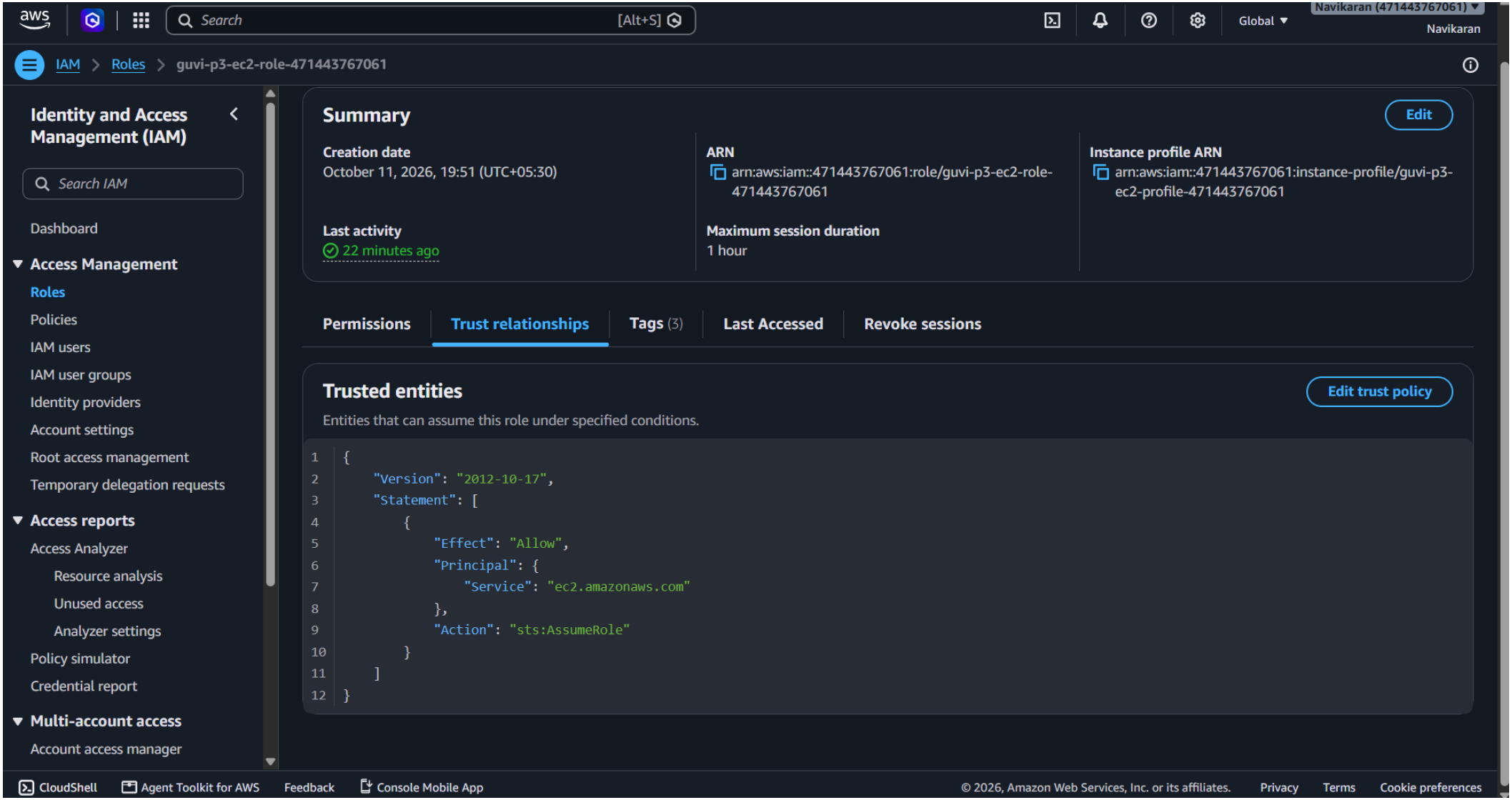
# EC2 IAM role permissions

The EC2 role has the customer inline policy named guvi-p3-app-s3-read; the role summary includes an instance profile ARN.



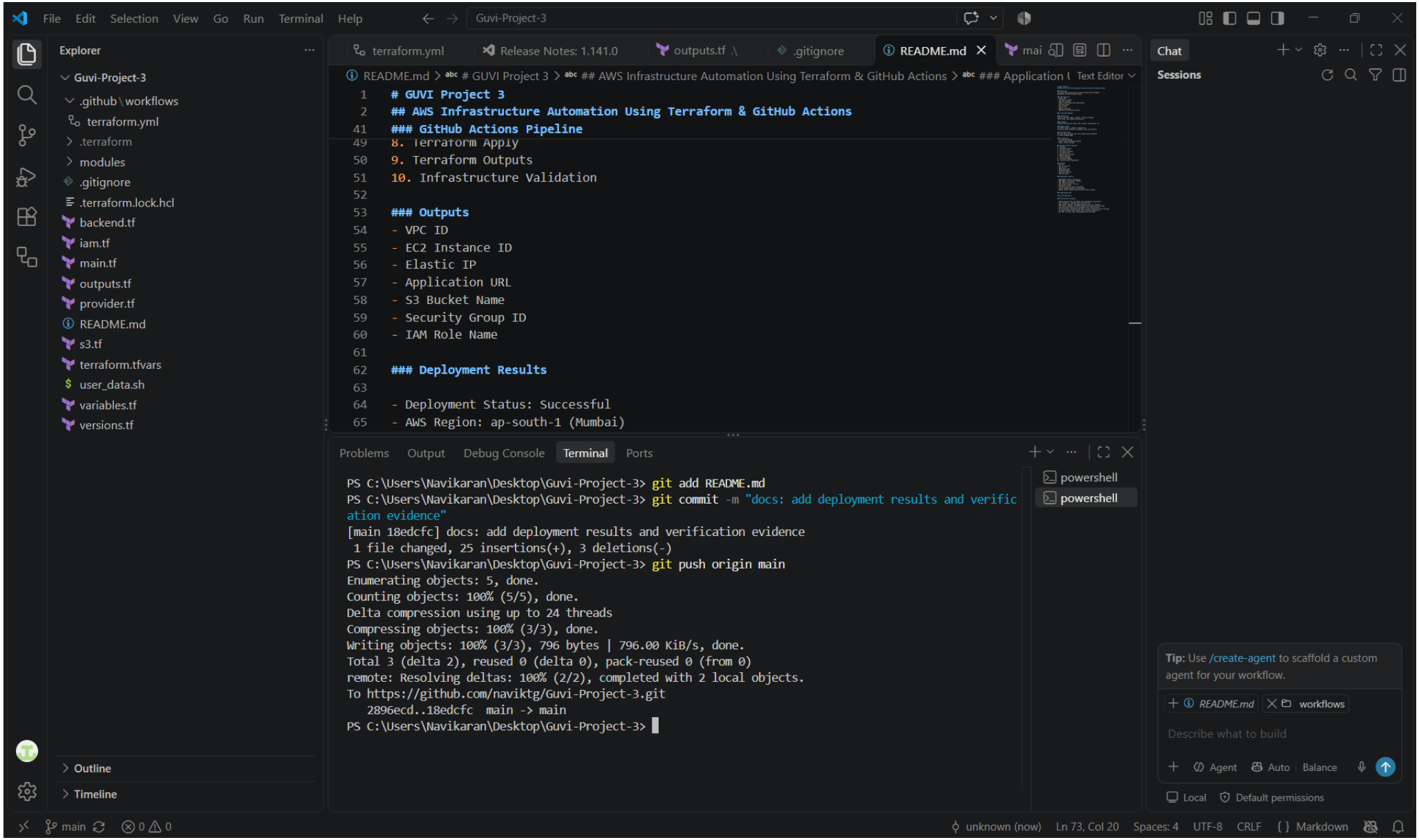
# EC2 role trust relationship

The trust policy allows the ec2.amazonaws.com service to assume the EC2 role.



# README update committed and pushed

The terminal shows the documentation commit and a successful push to the main branch.



# Final verification and operational notes

## Core deployment evidence: successful

The GitHub Actions run completed successfully. The website was opened in a browser, and the EC2, networking, Security Group, S3, remote-state, and EC2 IAM configurations were inspected through the AWS Console.

### After evidence capture: cost-control actions

- EC2 instances were stopped after the live-website and running-instance screenshots were captured.
- Elastic IP 13.205.240.3 was disassociated and released. The IP is historical evidence and may not be assigned to the account now.
- The GitHub repository and workflow history remain available: https://github.com/naviktg/Guvi-Project-3.
- The Terraform state bucket contains project3/terraform.tfstate. Keep it until you are certain no further Terraform operations are required.

**Security follow-up**

A previously exposed IAM access key was shown as Inactive in the later screenshot. Keep it inactive; if any copy of the secret was exposed, ensure it is deleted/rotated. The GitHub Actions role had broad AdministratorAccess during setup; for real production use, replace it with least-privilege permissions.